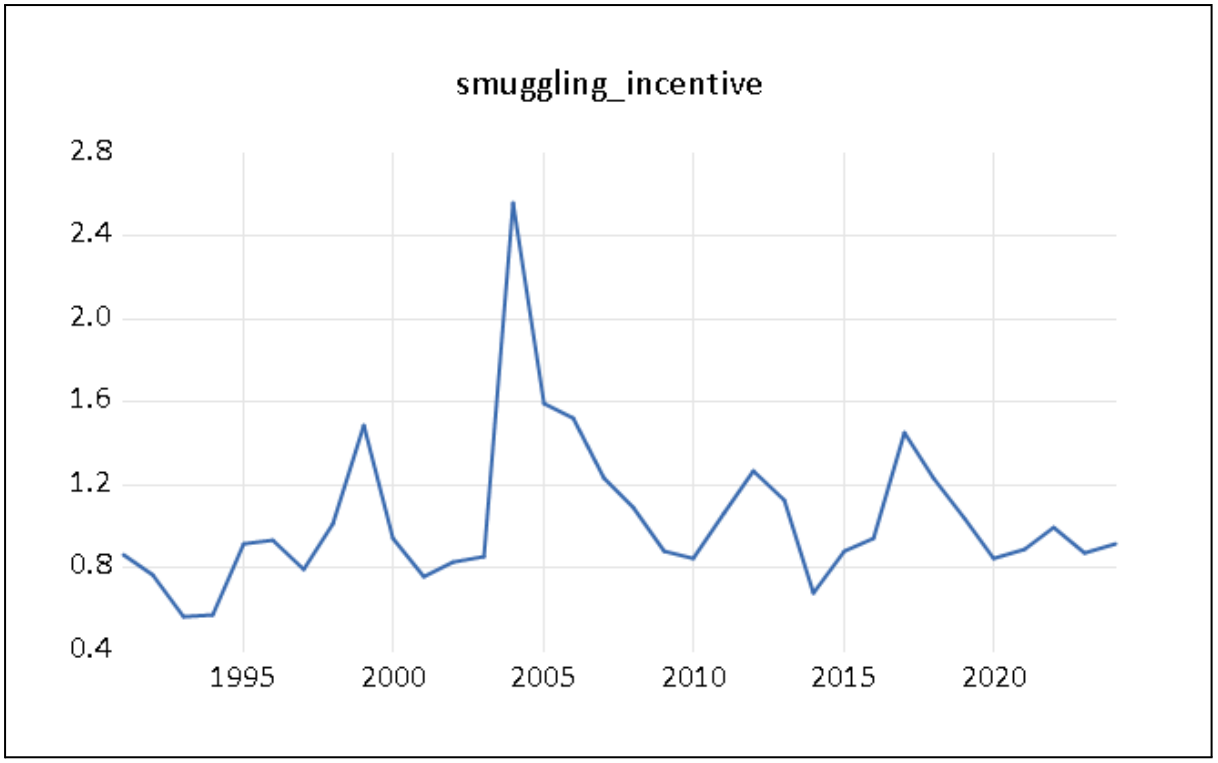

smuggling_incentive
2.8
2.4
2.0
1.6
1.2
0.8
0.4
1995
2000
2005
2010
2015
2020

Pruebo los test de Dickey-Fuller:

Null Hypothesis: SMUGGLING_INCENTIVE has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=8)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.672696	0.0387
Test critical values:	1% level	-4.262735	
	5% level	-3.552973	
	10% level	-3.209642	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(SMUGGLING_INCENTIVE)
Method: Least Squares
Date: 10/01/26 Time: 18:52
Sample (adjusted): 1992 2024
Included observations: 33 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
SMUGGLING_INCENTIVE(-1)	-0.622694	0.169547	-3.672696	0.0009
C	0.625826	0.205943	3.038831	0.0049
@TREND("1991")	0.001283	0.006556	0.195769	0.8461

R-squared	0.311184	Mean dependent var	0.001768
Adjusted R-squared	0.265262	S.D. dependent var	0.415255
S.E. of regression	0.355943	Akaike info criterion	0.858418
Sum squared resid	3.800873	Schwarz criterion	0.994465
Log likelihood	-11.16390	Hannan-Quinn criter.	0.904194
F-statistic	6.776482	Durbin-Watson stat	1.956305
Prob(F-statistic)	0.003729		

La tendencia determinista no es significativa por lo tanto pruebo solo con intercepto.

Null Hypothesis: SMUGGLING_INCENTIVE has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=8)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.734610	0.0080
Test critical values:	1% level	-3.646342	
	5% level	-2.954021	
	10% level	-2.615817	

*MacKinnon (1996) one-sided p-values.

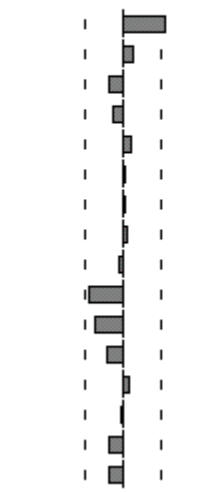
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(SMUGGLING_INCENTIVE)
Method: Least Squares
Date: 10/01/26 Time: 18:52
Sample (adjusted): 1992 2024
Included observations: 33 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
SMUGGLING_INCENTIVE(-1)	-0.618651	0.165653	-3.734610	0.0008
C	0.643452	0.182325	3.529137	0.0013

R-squared	0.310304	Mean dependent var	0.001768
Adjusted R-squared	0.288055	S.D. dependent var	0.415255
S.E. of regression	0.350379	Akaike info criterion	0.799089
Sum squared resid	3.805729	Schwarz criterion	0.889787
Log likelihood	-11.18497	Hannan-Quinn criter.	0.829606
F-statistic	13.94731	Durbin-Watson stat	1.961561
Prob(F-statistic)	0.000759		

El intercepto es significativo por lo que miro el estadístico tau al 5%.
Como -3.73<-2.95 rechazo la hipótesis nula de raíz unitaria al 5% de significatividad, por lo que no debo diferenciar la serie.

Date: 09/27/26 Time: 19:04
Sample: 1991 2024
Included observations: 34



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.380	0.380	5.3646	0.021
		2	0.093	-0.061	5.6943	0.058
		3	-0.127	-0.166	6.3304	0.097
		4	-0.096	0.020	6.7028	0.152
		5	0.076	0.149	6.9494	0.224
		6	0.012	-0.112	6.9553	0.325
		7	0.016	0.014	6.9671	0.432
		8	0.033	0.085	7.0180	0.535
		9	-0.038	-0.092	7.0905	0.628
		10	-0.299	-0.366	11.657	0.309
		11	-0.253	0.033	15.057	0.180
		12	-0.140	0.035	16.148	0.185
		13	0.064	-0.007	16.386	0.229
		14	-0.009	-0.182	16.391	0.290
		15	-0.122	-0.008	17.343	0.299
		16	-0.128	-0.022	18.463	0.298

Ya que tanto la Función de Autocorrelación como la Parcial son significativas en el primer rezago y se anulan en el segundo podría tratarse de una ARMA(1,1).

Dependent Variable: SMUGGLING_INCENTIVE
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/01/26 Time: 18:53
Sample (adjusted): 1992 2024
Included observations: 33 after adjustments
Failure to improve likelihood (non-zero gradients) after 18 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1991

Variable	Coefficient	Std. Error	t-Statistic	Prob.
@TREND	0.002338	0.010405	0.224721	0.8238
C	0.998236	0.206877	4.825261	0.0000
AR(1)	0.318043	0.464516	0.684676	0.4990
MA(1)	0.070640	0.487880	0.144790	0.8859

R-squared	0.148085	Mean dependent var	1.038999
Adjusted R-squared	0.059956	S.D. dependent var	0.373176
S.E. of regression	0.361816	Akaike info criterion	0.917853
Sum squared resid	3.796424	Schwarz criterion	1.099248
Log likelihood	-11.14458	Hannan-Quinn criter.	0.978887
F-statistic	1.680315	Durbin-Watson stat	1.985770
Prob(F-statistic)	0.193019		

Inverted AR Roots	.32
Inverted MA Roots	-.07

Saco el coeficiente de tendencia que no es significativo:

Dependent Variable: SMUGGLING_INCENTIVE
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/01/26 Time: 18:54
Sample (adjusted): 1992 2024
Included observations: 33 after adjustments
Failure to improve likelihood (non-zero gradients) after 16 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1991

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.039264	0.098295	10.57287	0.0000
AR(1)	0.334473	0.440770	0.758839	0.4539
MA(1)	0.056853	0.467965	0.121490	0.9041

R-squared	0.146637	Mean dependent var	1.038999
Adjusted R-squared	0.089746	S.D. dependent var	0.373176
S.E. of regression	0.356037	Akaike info criterion	0.858945
Sum squared resid	3.802875	Schwarz criterion	0.994991
Log likelihood	-11.17260	Hannan-Quinn criter.	0.904721
F-statistic	2.577516	Durbin-Watson stat	1.986163
Prob(F-statistic)	0.092684		

Inverted AR Roots	.33
Inverted MA Roots	-.06

Saco uno de los componentes ya que ninguno es significativo:

Dependent Variable: SMUGGLING_INCENTIVE
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/01/26 Time: 18:55
Sample: 1991 2024
Included observations: 34
Failure to improve likelihood (non-zero gradients) after 7 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1990

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.032474	0.079723	12.95070	0.0000
MA(1)	0.339018	0.166206	2.039750	0.0497

R-squared	0.130686	Mean dependent var	1.033716
Adjusted R-squared	0.103520	S.D. dependent var	0.368768
S.E. of regression	0.349159	Akaike info criterion	0.790443
Sum squared resid	3.901180	Schwarz criterion	0.880228
Log likelihood	-11.43752	Hannan-Quinn criter.	0.821062
F-statistic	4.810647	Durbin-Watson stat	1.891208
Prob(F-statistic)	0.035666		

Inverted MA Roots	-.34

Date: 10/01/26 Time: 18:56
Sample: 1991 2024
Q-statistic probabilities adjusted for 1 ARMA term



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.051	0.051	0.0951	
		2	0.123	0.121	0.6732	0.412
		3	-0.142	-0.156	1.4707	0.479
		4	-0.090	-0.093	1.8033	0.614
		5	0.116	0.172	2.3729	0.668
		6	-0.030	-0.048	2.4130	0.790
		7	0.021	-0.049	2.4323	0.876
		8	0.015	0.070	2.4422	0.931
		9	0.042	0.060	2.5275	0.960
		10	-0.282	-0.367	6.5732	0.681
		11	-0.131	-0.096	7.4877	0.679
		12	-0.141	0.024	8.5969	0.659
		13	0.117	0.063	9.3978	0.669
		14	-0.015	-0.163	9.4113	0.741
		15	-0.092	-0.064	9.9604	0.765
		16	-0.086	-0.008	10.462	0.790

Tanto el intercepto como el componente MA(1) dieron significativos y con ruido blanco, sin embargo debo probar con el componente AR(1):

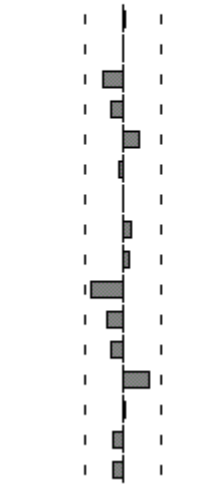
Dependent Variable: SMUGGLING_INCENTIVE
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/01/26 Time: 18:54
Sample (adjusted): 1992 2024
Included observations: 33 after adjustments
Convergence achieved after 3 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.040089	0.098594	10.54925	0.0000
AR(1)	0.381349	0.165653	2.302093	0.0282

R-squared	0.145997	Mean dependent var	1.038999
Adjusted R-squared	0.118448	S.D. dependent var	0.373176
S.E. of regression	0.350379	Akaike info criterion	0.799089
Sum squared resid	3.805729	Schwarz criterion	0.889787
Log likelihood	-11.18497	Hannan-Quinn criter.	0.829606
F-statistic	5.299630	Durbin-Watson stat	1.961561
Prob(F-statistic)	0.028210		

Inverted AR Roots	.38

Date: 10/01/26 Time: 18:55
Sample (adjusted): 1992 2024
Q-statistic probabilities adjusted for 1 ARMA term



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.013	0.013	0.0061	
		2	-0.005	-0.005	0.0069	0.934
		3	-0.185	-0.185	1.3211	0.517
		4	-0.104	-0.103	1.7536	0.625
		5	0.139	0.145	2.5503	0.636
		6	-0.036	-0.076	2.6059	0.760
		7	0.003	-0.039	2.6062	0.856
		8	0.069	0.123	2.8291	0.900
		9	0.055	0.066	2.9745	0.936
		10	-0.286	-0.364	7.0702	0.630
		11	-0.146	-0.105	8.1874	0.611
		12	-0.111	-0.031	8.8618	0.635
		13	0.223	0.112	11.740	0.467
		14	0.012	-0.154	11.749	0.548
		15	-0.088	-0.054	12.249	0.586
		16	-0.086	-0.025	12.751	0.622

Ambas estimaciones son significativas y con ruido blanco, por lo tanto debo usar el criterio de Schwarz para elegir entre ambos modelos, sin embargo para hacerlo ambas estimaciones deben tener la misma cantidad de observaciones, por lo tanto debo estimar nuevamente el modelo MA(1) con 33 observaciones como tiene el AR(1), en vez de 34:

Dependent Variable: SMUGGLING_INCENTIVE
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/01/26 Time: 18:57
Sample: 1992 2024
Included observations: 33
Failure to improve likelihood (non-zero gradients) after 7 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1991

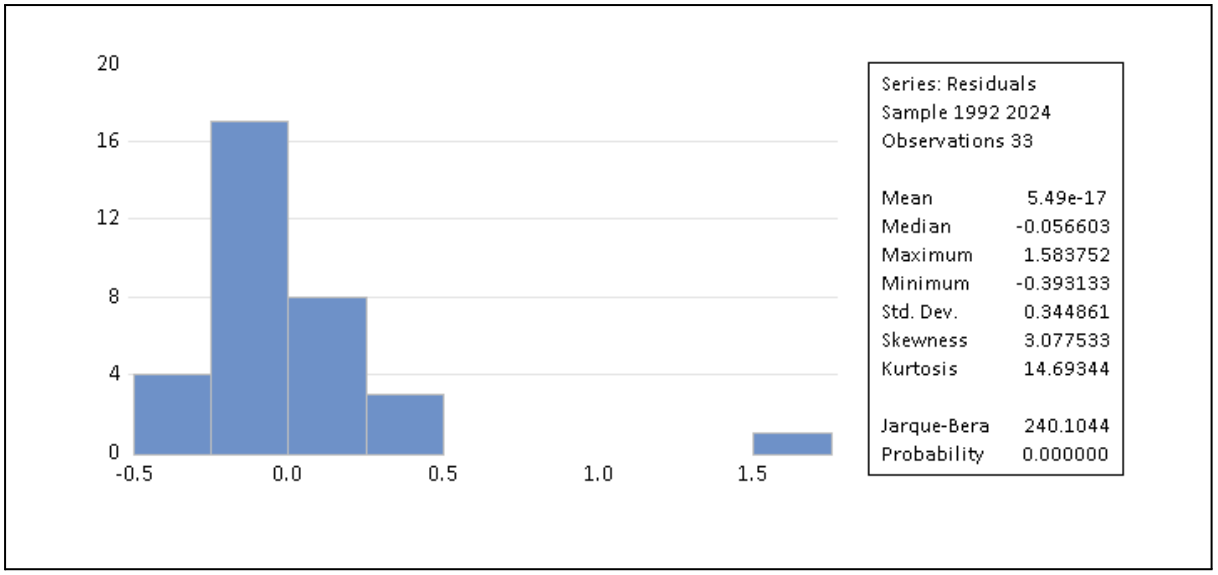
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.037417	0.081930	12.66219	0.0000
MA(1)	0.337279	0.168908	1.996820	0.0547

R-squared	0.128138	Mean dependent var	1.038999
Adjusted R-squared	0.100013	S.D. dependent var	0.373176
S.E. of regression	0.354024	Akaike info criterion	0.819786
Sum squared resid	3.885316	Schwarz criterion	0.910483
Log likelihood	-11.52647	Hannan-Quinn criter.	0.850303
F-statistic	4.556067	Durbin-Watson stat	1.890656
Prob(F-statistic)	0.040817		

Inverted MA Roots	-.34

Al utilizar una observación menos, debido a que 33 no son muchas observaciones, el componente MA(1), que antes tenía un p-valor de 0.0497, que estaba muy cerca de la zona de aceptación al 5% paso a dejar de ser significativo para un α=5%.

- Por lo tanto el modelo elegido es el AR(1) con intercepto y sin tendencia.



El test de Jarque-Bera de normalidad de residuos rechaza rotundamente que los residuos se comportan normalmente.